

Integrated e-Filing & Centralized Processing Center

Standard Operating Procedure ERI Type 3 Onboarding document

IEC2.0 Programme | ERI Operations

Revision history

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Table of Contents

1. What is a Type 3 ERI?	4
2. Scope of a Type 3 ERI.....	4
3. Steps for Integration & UAT Sanity Testing.....	4
4. Actions Required from Type 3 ERIs for Every Assessment Year (AY)	4
5. Digest Generation Guide for ERI Users –	5
5.1 What is the Digest Value?	5
5.3. How is the Digest Generated? (Step-by-Step).....	5
5.4 Visual Diagram of Digest Generation Process	6
5.5 Key Inputs Required.....	6
5.6 Best Practices for ERI Users	6
5.7 Contact for Support.....	6

1. What is a Type 3 ERI?

A Type 3 ERI (e-Return Intermediary) is an organisation authorised by the Income Tax Department (ITD) to build a schema-based offline utility that assists taxpayers in preparing their Income Tax Returns (ITRs). Type 3 ERIs do not receive any API access. Their responsibility is to generate valid ITR JSON files strictly based on the latest schemas published on the ITD portal, which taxpayers later upload using Offline Mode.

2. Scope of a Type 3 ERI

- JSON generation based on schemas published on the ITD portal.
- No access to e-Filing APIs, validation APIs, or pre-fill APIs.
- Development of a software utility to collect and validate taxpayer data as per schema.
- Taxpayer uploads the JSON in Offline Mode on the ITD portal.
- Ensure annual updates and compliance with ITD schema changes.

3. Steps for Integration & UAT Sanity Testing

Step 1: Request UAT credentials by emailing erihelp@incometax.gov.in. The ERI Technical Operation Team provides UAT credentials and test data.

Step 2: Generate UAT JSONs based on latest schemas using UAT credentials and provided test data.

Step 3: Submit JSONs for validation for each ITR you intend to support.

Step 4: ERI Technical Operation Team performs sanity checks on the JSON generated with UAT Credentials (schema, structure, mandatory fields). Turnaround: 1–2 working days.

Step 5: After successful testing and ITD approval, ERI Technical Operation Team enables the SW_ID for the specific ITR and AY. ERI users will be informed post enablement through email from erihelp@incometax.gov.in

4. Actions Required from Type 3 ERIs for Every Assessment Year (AY)

- Download latest schemas as soon as they are published.
- Update your utility to comply with new AY schemas.
- Proactively generate UAT JSONs using UAT credentials and new test data. (UAT credentials are ERI_ID, SW_ID, Secret Key & Iteration).
- Submit to ERI Technical Operation Team for UAT sanity by sharing the JSONs through erihelp@incometax.gov.in
- After approval, SW_ID is enabled for the specific ITR and AY (no additional action required).
- Perform internal QA for accuracy and schema alignment.

Please note, Enabling ITRs is purely voluntary process from the ERI users. ERI Users may choose the ITRs, based on their business requirements. It's not mandatory that every ERI user must enable all ITRs for given AY.

5. Digest Generation Guide for ERI Users –

This section explains what the “Digest” value is, why it is important, and how it is generated in the Income Tax Return (ITR) JSON file. It is written for ERI users and stakeholders their reference.

5.1 What is the Digest Value?

The Digest is a unique security code embedded in the ITR JSON file. It ensures the integrity of the data by confirming that the file has not been altered after generation think of it as a digital fingerprint for your ITR file.

5.2. Why is the Digest Important?

It guarantees data authenticity, prevents tampering or unauthorized changes.
It is required for validation during submission.

5.3. How is the Digest Generated? (Step-by-Step)

Step 1: Read the Input JSON

Step 2: Minify the JSON

Step 3: Locate Digest Field, Find the position of the "Digest" field in the JSON.

Replace the value with a placeholder (e.g., "-").

Step 4: Load the Secret key and Iteration

Step 5: Generate New Digest, Using HMAC-SHA256 algorithm:

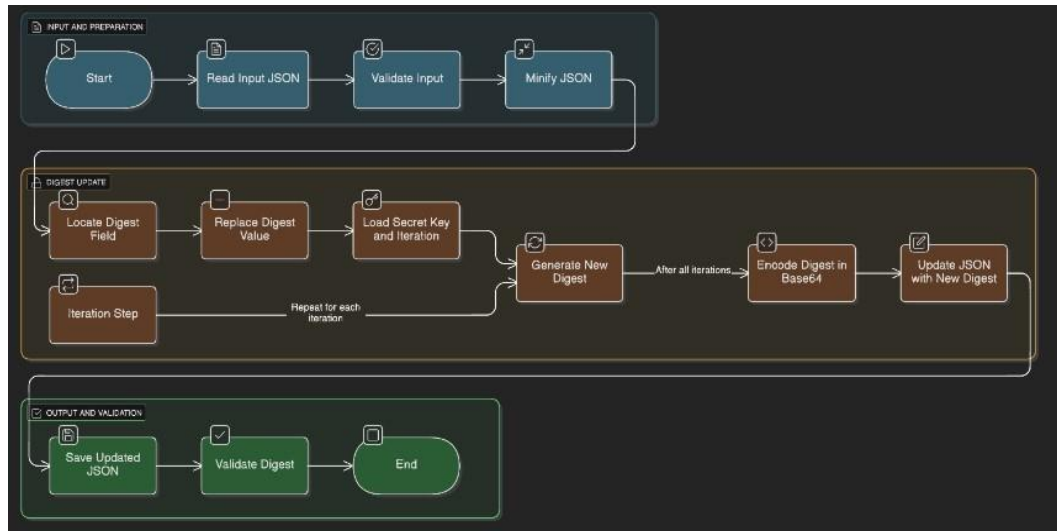
1. Initialize with the secret key.
2. Hash the modified JSON string.
3. Repeat hashing for the specified number of iterations (Iteration value provided with credentials).
4. Encode the final hash in Base64 format.

Step 6: Update JSON

Replace the placeholder with the newly generated digest.

Save the updated JSON to a new file.

5.4 Visual Diagram of Digest Generation Process



5.5 Key Inputs Required

ITR JSON file with fields like Digest and SWCreatedBy.

SWCreatedBy code (e.g., "SW20013802") – identifies the ERI system.

Secret key & iteration count – predefined for each ERI code (managed securely).

5.6 Best Practices for ERI Users

Do not manually edit the Digest value.

Ensure the JSON file structure remains unchanged—spacing and formatting can affect the Digest.
Remove all extra spaces in the JSON that are not part of variable values.

Always use the official utility or API provided for Digest generation.

Keep your secret keys confidential and follow organizational security guidelines.

5.7 Contact for Support

For assistance, please reach out to ERI Helpdesk at: erihelp@incometax.gov.in.